Sujal Sakpal

Roll No.: 231060064
B.Tech Electronics Engineering
Veermata Jijabai Technological Institute, Mumbai

+91-8767007296
sakpalsujal121@gmail.com
sasakpal$_b$23@*ee.vjti.ac.in*
GitHub
LinkedIn

Education

•Veermata Jijabai Technological Institute (VJTI) *2027*
BTech Electronics Engineering CGPA : 7.96

•NES Junior College *2023*
State Board of Secondary and Higher Secondary Education, Maharashtra Percentage: 88.17

Personal Projects

•CodeGen | Agentic AI Platform for Competitive Programming *Oct, 2024*

An AI-driven web platform that generates multiple solution strategies for competitive programming problems using autonomous agents, knowledge graphs, and retrieval-augmented generation.

- Tools & technologies used: CrewAI, LangChain, Neo4j, FAISS, FastAPI, Python, HTML, CSS, JavaScript
- Curated problem data and editorial strategies from Striver's DSA Sheet to build a rich dataset for solution generation.
- Used CrewAI to orchestrate reasoning agents; integrated Neo4j for structured knowledge graphs and FAISS for semantic retrieval in a RAG pipeline.
- Developed a full-stack web service using FastAPI and Python for the backend, with a responsive frontend built in HTML, CSS, and JavaScript.

•Grammarator | T5-based AI Grammar Correction Tool *Aug, 2024*

An NLP tool that performs grammar correction using fine-tuned T5 models, built with a focus on accuracy, efficiency, and real-world usability.

- Tools & technologies used: HuggingFace Transformers, Flan-T5, LoRA, PyTorch, JFLEG Dataset, Python
- Fine-tuned `flan-t5-large` on the JFLEG dataset using parameter-efficient LoRA training to enhance grammar correction accuracy.
- Built a stepwise validation pipeline to inspect model outputs, and planned extensions like misinterpretation detection and custom style correction.

•Event Management Website | End-to-End Event Platform *Dec, 2024*

A full-stack event management platform enabling event creation, ticket sales, payment handling, user authentication, and file uploads with modern web technologies.

- Tools & technologies used: Next.js, Tailwind CSS, Stripe, Clerk, UploadThings, MongoDB
- Built a responsive and scalable web platform where organizers can create events, upload banners/files, and manage ticket sales in real time.
- Integrated `Stripe` for secure payment processing and `Clerk` for user authentication and role-based access control.
- Utilized `MongoDB` for storing dynamic event and user data, with seamless file uploads handled via `UploadThings` for media-rich event pages.

•Movie Recommender Website | Content-Based Recommendation System *Sept, 2024*

A personalized movie recommendation system leveraging content similarity metrics and classic ML techniques for fast and accurate suggestions.

- Tools & technologies used: NumPy, Pandas, Scikit-learn, Cosine Similarity, FastAPI, HTML, CSS, JavaScript
- Implemented a content-based recommendation engine using TF-IDF and cosine similarity on movie metadata to suggest similar films.
- Developed the backend with FastAPI for high-performance API endpoints and built a clean, responsive frontend using HTML, CSS, and JavaScript.

Positions of Responsibility

•Department Head, Enthusia - Sports Committee *Aug 2024 - March 2025*

•Marketing Executive, Ecell - Entrepreneurship Cell *Aug 2023 - March 2024*

Achievements

•First Place in College, 12th Grade Secured first rank in my junior college.

•Second Place in School, 10th Grade Achieved second place in my school